

titanic

Predicting **survived** · classification · generated 28 July 2026, 06:22 UTC

STRONG

This model is reliable enough to act on.

SUMMARY

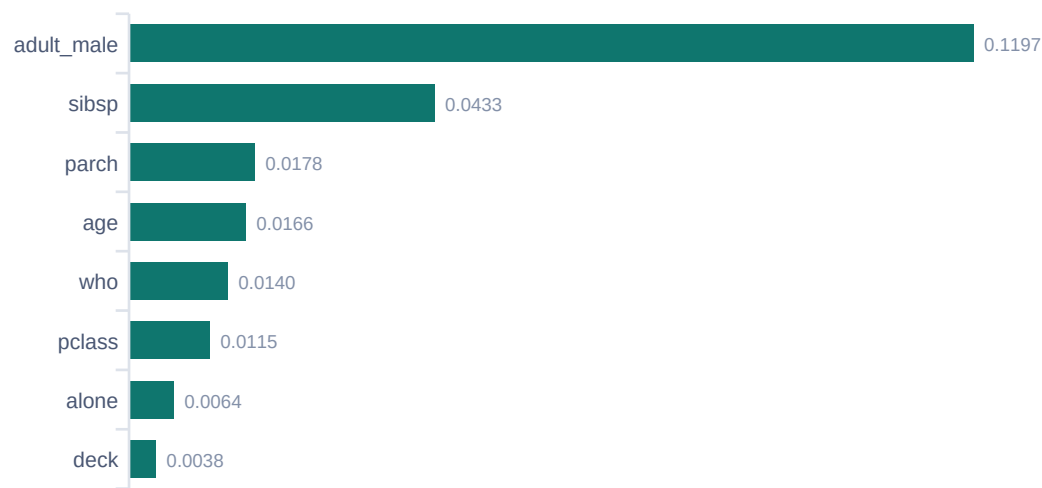
The target column 'survived' was chosen for this classification model, and the best performing model was logistic_regression with an accuracy of 0.8471 and an roc_auc of 0.9009. The features that actually drove the predictions were "adult_male", "sibsp", "parch", "age", and "who", with "adult_male" being the strongest contributor at 0.1197. The model's performance is notable, although it is worth observing that the data used for training contained 107 fully duplicated rows and 19.9% missing values in the 'age' column. Alternative models, such as random_forest, were also tried, but logistic_regression performed slightly better with an f1 score of 0.8475 compared to random_forest's 0.8318.

COLUMNS REMOVED BEFORE MODELLING

These columns restated the answer. Left in, they would have produced a perfect score and taught the model nothing. Every figure in this report comes from a model trained without them.

Column	Why
alive	knowing 'alive' determines the target (mutual information 1.000); it restates the answer

WHAT DROVE THE PREDICTIONS



Mean decrease in score when each column is shuffled, over 5 repeats.

HOW FAR TO TRUST THIS

Check	Result	Detail
class separation	pass	ROC-AUC 0.9009 meets the 0.65 threshold.

f1 threshold	pass	F1 0.8475 meets the 0.6 threshold.
target leakage	found	Removed before training because they restate the target: alive (mutual information 1.0)
target is derived	pass	The target is not a combination of its own features.
sample size	pass	627 training rows is sufficient; sample size is NOT a problem.

Contributed almost nothing: deck, sex, fare, embarked, class, embark_town

HOW THIS WAS PRODUCED

Cleaning applied first:

Action	Rows	Columns
drop duplicate rows	107	—
Rows used	627 train / 157 test	
Features	13	
Columns excluded	1	
Best model	logistic_regression	

Models compared:

Model	Score	Seconds
logistic_regression	f1 0.8475	0.40 chosen
random_forest	f1 0.8318	2.27

Why it stopped:

Quality check passed (roc_auc=0.9009).

LIMITS OF THIS ANALYSIS

— Feature importance shows association, not cause. A column that predicts an outcome does not necessarily influence it.

— Scores are measured on data held back from training, but they describe this dataset. Performance on new data may differ.